

▷ One branch

The simplest branching command is an `if` statement. It only has one branch, which the computer takes if the condition is `True`. This program asks the user to say if it's dark outside. If it is, the program pretends that the computer is going to sleep! If it's not dark, `is_dark == 'y'` is `False`, so the "Goodnight!" message isn't displayed.

This line asks the user to reply "y" (yes) or "n" (no).

```
is_dark = input('Is it dark outside? y/n')
if is_dark == 'y':
    print('Goodnight! Zzzzzzzzzzzzzzzzzzzzz...')
```

Condition

This branch is taken if the condition is `True`.

The code shows this message in the shell window.

▷ Two branches

Do you want a program to do one thing if a condition's `True` and another thing if it's `False`? If so, you need a command with two branches, called an `if-else` statement. This program asks if the user has tentacles. If they answer "Yes", it decides they must be an octopus! If they answer "No", it decides they're human. Each decision prints a different message.

This line asks for input from the user.

Condition

```
tentacles = input('Do you have tentacles? (n/y)')
if tentacles == 'y':
    print('I never knew octopuses could type!')
else:
    print('Greetings, human!')
```

This block runs if the condition is `True`.

This block runs if the condition is `False`.

▷ Multiple branches

When there are more than two possible paths, the statement `elif` (short for "else-if") comes in handy. This program asks the user to type in the weather forecast: either "rain", "snow", or "sun". It then chooses one of three branches and weather conditions.

```
weather = input('What is the forecast for today? (rain/snow/sun)')

if weather == 'rain':
    print('Remember your umbrella!')
elif weather == 'snow':
    print('Remember your woolly gloves!')
else:
    print('Remember your sunglasses!')
```

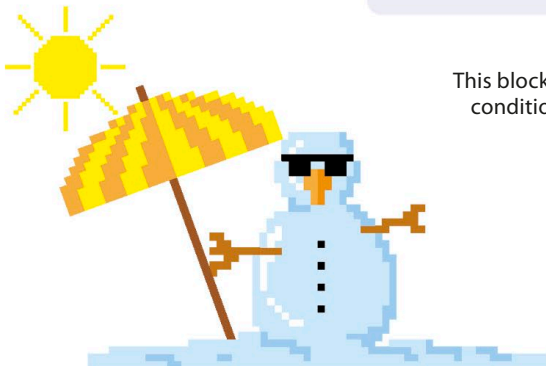
First condition

This block runs if the first condition is `True`.

Second condition

This block runs if the second condition is `True`.

This block runs if both conditions are `False`.



△ How it works

An `elif` statement must always come after `if` and before `else`. In this code, `elif` checks for snow only when the condition set by the `if` statement is `False`. You could insert additional `elif` statements to check for more types of weather.